

17	A sound wave of frequency 400 Hz is travelling in air at a speed of 320 m s⁻¹. What is the difference in phase between two points on the wave 0.2 m apart in the direction of travel?			
	A $\frac{\pi}{4}$ rad	B $\frac{\pi}{2}$ rad	C $\frac{2\pi}{5}$ rad	D $\frac{4\pi}{5}$ rad
	Answer: B $\lambda = \frac{v}{f}$ $= \frac{320}{400}$ $= 0.8 \text{ m}$ $\frac{\Delta\phi}{2\pi} = \frac{\Delta x}{\lambda}$			

$$= 2\pi \left(\frac{0.2}{0.8} \right)$$

$$= \frac{\pi}{2} \text{ rad}$$